# Automatic Segmentation of Multiple Sclerosis Lesions Using a 2.5D U-Net

**Course:** Deep Learning **Dataset:** MSLesSeg (Guarnera et al., *Scientific Data*, 2025)
**Framework:** PyTorch **Runtime Environment:** Google Colab (Tesla T4 GPU, 15.6 GB VRAM)

## 1. Introduction

Multiple Sclerosis (MS) is a chronic inflammatory disorder of the central nervous system characterised by demyelination of the cerebral white matter. Magnetic Resonance Imaging (MRI) is the primary tool for diagnosis and disease monitoring; in particular, **FLAIR** (Fluid Attenuated Inversion Recovery) sequences allow MS lesions to be visualised as hyperintense areas that are clearly distinguishable from the surrounding tissue.

Manual lesion segmentation is a time-consuming, subjective, and poorly reproducible process. The goal of this project is to develop an **automatic segmentation system** based on deep learning, using a U-Net architecture adapted to the volumetric nature of MRI data through a so-called **2.5D approach**.

## 2. Dataset

The dataset used is **MSLesSeg**, containing 115 FLAIR volumes from 75 patients:

| Split | Patients | Folder Structure |
|---|---|---|
| Training | 53 | `train/P{id}/T{tp}/` (with timepoint subfolders) |
| Test | 22 | `test/P{id}/` (files directly in the patient folder) |

Each volume is accompanied by a manually annotated **binary lesion mask**. The training and test sets consist of **entirely separate patients**, ensuring no data leakage.

## 3.1 Attempts

## Attempt 1 2D U-Net with Lesion Size Threshold Filtering

> **Architecture:** Standard 2D U-Net processing one axial FLAIR slice at a time.
>
> **Key design choice lesion threshold filtering:** Only slices containing lesions above a minimum pixel-area threshold were included. This filter was applied to **both** the training/validation set and the **test set**, meaning that slices with very small or absent lesions were excluded from evaluation entirely.

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**Result:** Dice score of **0.75**.

**Why this score is misleading:** Although the Dice score appeared high, this approach introduced a form of data leakage in the evaluation: by filtering out difficult cases (slices with tiny or no lesions) from the test set, the model was only evaluated on the "easy" subset of the data. The model never learned to handle lesion-free or near-lesion-free slices, making it unsuitable for clinical use.

## Attempt 2: 2.5D U-Net, Split by Slice, No Data Augmentation

**Architecture:** 2.5D U-Net

**Key design changes:** - The train/validation split was performed **at the slice level** rather than at the patient level. This means slices from the same patient could appear in both the training and the validation sets. - No data augmentation was applied. - No lesion-size threshold filtering was applied to the test set.

**Result:** Dice score of **0.65**.

## Attempt 3 2.5D U-Net with Patient-Level Split and Data Augmentation

The one that will be discussed in this report.

## 3.2 The 2.5D Approach: Motivation

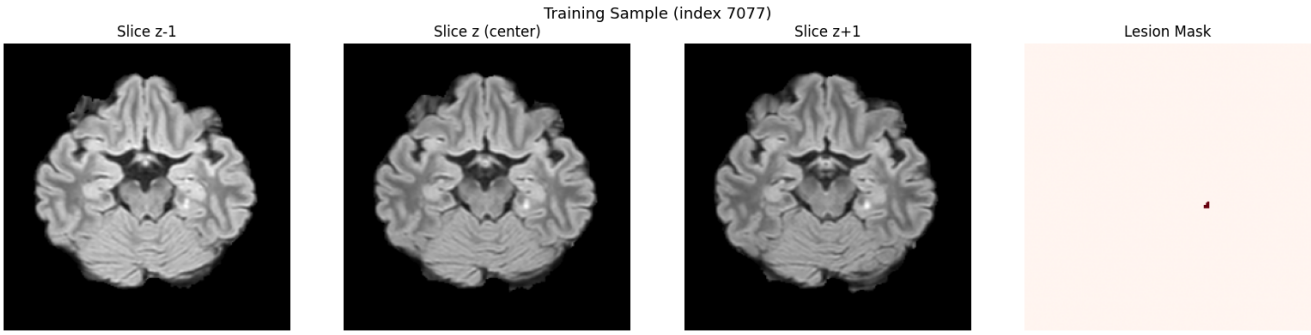
MRI scans are inherently 3D volumes. There are three main strategies for processing them with convolutional networks:

| Approach | Description | Pros | Cons |
|---|---|---|---|
| **2D** | Each slice processed independently | Lightweight, simple | Loses inter-slice context |
| **3D** | 3D convolutions on the full volume | Best spatial context | Very high memory cost |
| **2.5D** | Adjacent slices stacked as input channels | Good balance of context and efficiency | Slightly less context than 3D |

The **2.5D** approach was chosen for three reasons:

1. **Memory constraints:** a full 3D U-Net requires far more VRAM than is available on the Colab free tier (~15 GB).
2. **Spatial context:** unlike pure 2D, the model can observe neighbouring slices, capturing the three-dimensional extent of lesions.
3. **Implementation simplicity:** the architecture uses standard 2D convolutions, making it much easier to implement and debug.

In practice, for each axial slice at index *z*, the **3 adjacent slices** [z-1, z, z+1] are stacked as input channels (analogous to the RGB channels of a natural image). The output is the segmentation mask for the **central slice only**.

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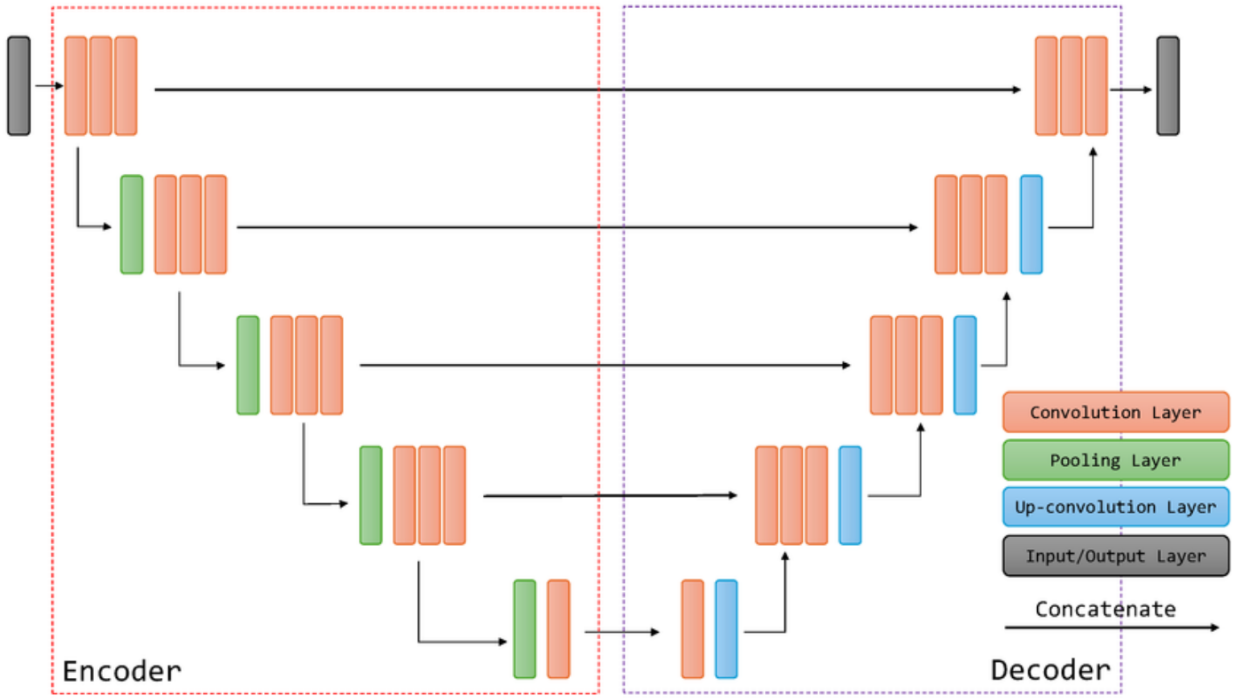
## 4. Model Architecture

The model follows the classic **U-Net** design (Ronneberger et al., 2015) with an encoder-decoder structure and skip connections:




https://mask2former.com/how-does-mask2former-compare-with-u-net/

**Building block (DoubleConv):** each encoder and decoder level consists of two 3x3 convolutions, each followed by Batch Normalisation and ReLU.

**Model parameters:** - Base features: 32 - Total parameters: **7,765,985** (~7.8M) - Size on disk: ~31.1 MB (float32)

The choice of 32 base features (instead of 64 in the original U-Net) keeps the model lightweight and compatible with the available resources.

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# 5. Data Preparation

## 5.1 Pre-processing

For each MRI volume: 1. **Loading** the FLAIR volume and lesion mask in NIfTI format (.nii.gz) via the `nibabel` library. 2. **Intensity normalisation** of FLAIR values to the [0, 1] range using min-max scaling. 3. **Resizing** each axial slice to a target resolution of **256 x 256** pixels, using bilinear interpolation for FLAIR images and nearest-neighbour interpolation for masks (to preserve binary values). 4. **Preloading into RAM** of the entire dataset to avoid repeated reads from Google Drive during training.

## 5.2 Construction of 2.5D Samples

For each slice at index *z* (excluding border slices of the volume): - **Input:** slices [z-1, z, z+1] stacked into a tensor of shape (3, 256, 256). - **Target:** the binary mask of slice *z*, as a tensor of shape (1, 256, 256).

## 5.3 Sample Balancing

- **All** slices containing at least one lesion pixel are included.
- **15%** of lesion-free slices are retained as negative examples, so the model also learns to recognise healthy tissue.

## Data Augmentation (Training Only)

To improve generalisation, the following augmentations are applied on-the-fly to training slices (each with probability 0.5):

| Type | Augmentation |
|---|---|
| Spatial | Horizontal flip |
| Spatial | Vertical flip |
| Spatial | Random 90° rotation (0°, 90°, 180°, 270°) |
| Spatial | Small random rotation (−15° to +15°) |
| Intensity | Random brightness shift (±0.1) |
| Intensity | Random contrast scaling (factor 0.8–1.2) |
| Intensity | Gaussian noise ($\sigma = 0.02$) |

Spatial augmentations are applied identically to both the input and the mask; intensity augmentations are applied to the input only.

# 6. Training

## 6.1 Configuration

| Hyperparameter | Value |
|---|---|
| Loss function | Binary Cross-Entropy (BCE) |
| Optimiser | Adam |
| Learning rate | 1 x 10^-4 |
| Batch size | 16 |
| Epochs | 20 |
| Train/val split | 80% / 20% (random, seed=5) |

## 6.2 Validation Split

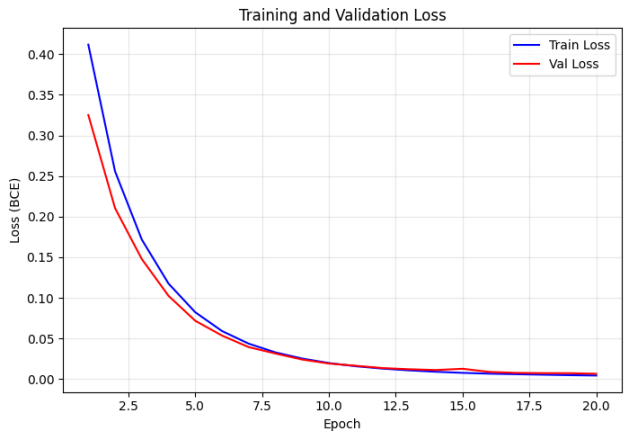
Total training + validations patients: 53

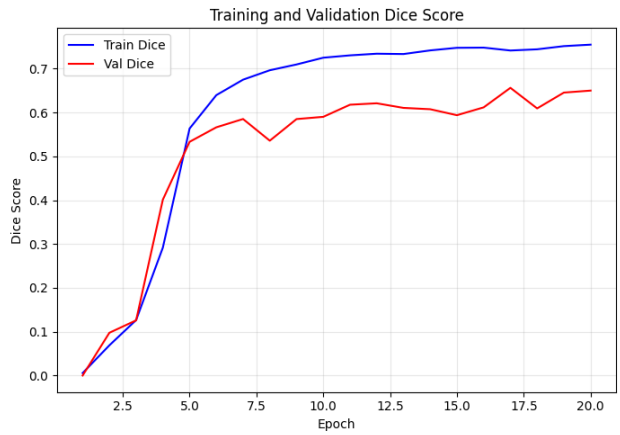
**Train patients** (42): ['P1', 'P10', 'P11', 'P13', 'P15', 'P16', 'P17', 'P18', 'P19', 'P2', 'P22', 'P23', 'P24', 'P26', 'P28', 'P29', 'P3', 'P30', 'P31', 'P32', 'P33', 'P34', 'P35', 'P36', 'P37', 'P38', 'P4', 'P40', 'P41', 'P42', 'P43', 'P44', 'P45', 'P46', 'P48', 'P5', 'P50', 'P51', 'P53', 'P6', 'P8', 'P9']

**Val patients** (11): ['P12', 'P14', 'P20', 'P21', 'P25', 'P27', 'P39', 'P47', 'P49', 'P52', 'P7']

## 6.3 Training Progress

| Epoch | Train Loss | Train Dice | Val Loss | Val Dice |
|---|---|---|---|---|
| 1 | 0.4117 | 0.0058 | 0.3251 | 0.0002 |
| 5 | 0.0823 | 0.5634 | 0.0718 | 0.5331 |
| 10 | 0.0197 | 0.7249 | 0.0191 | 0.5901 |
| 15 | 0.0591 | 0.7475 | 0.0127 | 0.5938 |
| **17** | 0.0060 | 0.7413 | 0.0076 | **0.6562** |
| 20 | 0.0044 | 0.7547 | 0.0065 | 0.6499 |




# 7. Test Set Evaluation

## 7.1 Metrics

- **Dice Score (F1):** measures the overlap between prediction and ground truth.

  Dice = 2|P ∩ G| / (|P| + |G|)

- **IoU (Jaccard):** ratio of intersection to union.

  IoU = |P ∩ G| / |P ∪ G|

- **Precision:** proportion of predicted lesion pixels that are lesion.

  Precision = TP / (TP + FP)

- **Recall (Sensitivity):** proportion of true lesion pixels that are correctly identified.

  Recall = TP / (TP + FN)

Metrics are computed **per patient** and then averaged, so that each patient contributes equally regardless of the number of slices.

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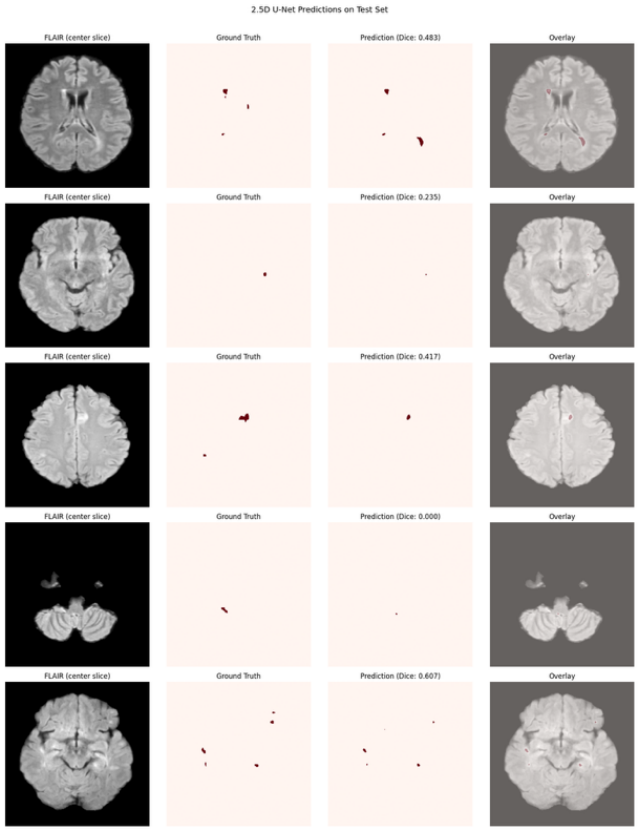
## 7.2 Per-Patient Results

| Patient | Dice | IoU | Precision | Recall | Slices |
|---|---|---|---|---|---|
| P54 | 0.7241 | 0.5675 | 0.5964 | 0.9212 | 65 |
| P55 | 0.6383 | 0.4687 | 0.7916 | 0.5347 | 106 |
| P56 | 0.7349 | 0.5809 | 0.9028 | 0.6196 | 54 |
| P57 | 0.7357 | 0.5820 | 0.8650 | 0.6401 | 135 |
| P58 | 0.6104 | 0.4393 | 0.6650 | 0.5641 | 109 |
| P59 | 0.4062 | 0.2549 | 0.3945 | 0.4186 | 66 |
| P60 | 0.6693 | 0.5030 | 0.6069 | 0.7461 | 115 |
| P61 | 0.6691 | 0.5027 | 0.5572 | 0.8371 | 88 |
| P62 | 0.6890 | 0.5255 | 0.8049 | 0.6023 | 102 |
| P63 | 0.7169 | 0.5588 | 0.6442 | 0.8081 | 81 |
| P64 | 0.7008 | 0.5395 | 0.6485 | 0.7624 | 100 |
| P65 | 0.7632 | 0.6170 | 0.7901 | 0.7380 | 124 |
| P66 | 0.7096 | 0.5499 | 0.6918 | 0.7282 | 108 |
| P67 | 0.7269 | 0.5709 | 0.8000 | 0.6660 | 126 |
| P68 | 0.3122 | 0.1850 | 0.6174 | 0.2089 | 71 |
| P69 | 0.8075 | 0.6772 | 0.8555 | 0.7647 | 83 |
| P70 | 0.6317 | 0.4617 | 0.4825 | 0.9146 | 87 |
| P71 | 0.5732 | 0.4018 | 0.4575 | 0.7672 | 80 |
| P72 | 0.7866 | 0.6483 | 0.8908 | 0.7042 | 95 |
| P73 | 0.6858 | 0.5219 | 0.8026 | 0.5987 | 96 |
| P74 | 0.6517 | 0.4833 | 0.6614 | 0.6422 | 88 |
| P75 | 0.7027 | 0.5416 | 0.8336 | 0.6073 | 112 |
| MEAN | 0.6657 | 0.5082 | 0.6982 | 0.6725 | |

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## 7.3 Analysis of Results

| Metric | Value |
|---|---|
| Best Validation Dice | **0.6562** (epoch 17) |
| **Test Dice (per-patient avg)** | **0.6657** |
| Test IoU | 0.5082 |
| Test Precision | 0.6982 |
| Test Recall | 0.6725 |
| Generalisation gap | 0.0095 → *good generalisation* |

Per-patient test Dice scores ranged from 0.31 (P68) to 0.81 (P69), reflecting the natural variability in lesion burden across patients.

### Why This Approach is More Reliable

1. **No data leakage** patient-level splitting ensures the validation and test scores reflect performance on truly unseen individuals.
2. **No artificial filtering of the test set** the model is evaluated on all patients, including those with small or sparse lesions.
3. **2.5D context** neighbouring slices provide continuity information that a purely 2D model lacks, helping the model capture the volumetric extent of lesions.
4. **Data augmentation** reduces overfitting and improves robustness to acquisition variability.

**Precision vs Recall:** on average, Precision (0.70) and Recall (0.67) are reasonably balanced, but with notable variability: - P68 has high precision but very low recall: the model is conservative, detecting few lesions but with high confidence. - P70 has low precision but very high recall: the model over-segments, producing many false positives.

## Comparison of Approaches

| | Attempt 1 | Attempt 2 | Attempt 3 |
|---|---|---|---|
| Architecture | 2D U-Net | 2D U-Net | **2.5D U-Net** |
| Train/val split | By slice | By slice | **By patient** |
| Test set filtering | Yes (threshold) | No | No |
| Data augmentation | No | No | **Yes** |
| Test Dice | 0.75 *(inflated)* | 0.65 | **0.6657** |

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| | | | |
|---|---|---|---|
| Evaluation fairness | Low | Medium | **High** |

The final 2.5D model achieves a lower raw Dice score than Attempt 1, but it does so under a strict and fair evaluation protocol. The 0.6657 test Dice is a trustworthy estimate of how the model would perform on new, unseen patients.

# 8. Discussion

## 8.1 Strengths

1. **Effective 2.5D approach:** the model leverages inter-slice context without the computational cost of 3D convolutions, making it trainable on limited hardware (Colab free tier).
2. **Simple and interpretable architecture**
3. **No lesion filtering:** including all lesions regardless of size makes the model more clinically realistic.
4. **Rigorous train/test separation:** no patient appears in both sets, ensuring a fair evaluation.

## 8.2 Limitations

1. **Context limited to 3 slices:** with only 3 adjacent slices, the model has a limited view along the axial direction. Increasing to 5 or 7 slices could improve segmentation consistency.

# 9. Conclusions

This project implemented a **2.5D U-Net** for automatic segmentation of Multiple Sclerosis lesions from FLAIR MRI scans, achieving a **mean Dice score of 0.67** on the test set. The model demonstrates the ability to identify lesions in the majority of cases, with performance varying considerably depending on the complexity of each patient.

The 2.5D approach proved to be an effective compromise between pure 2D and full 3D processing, enabling training on limited computational resources while retaining a degree of spatial context.

The main avenues for improvement lie in reducing overfitting and regularisation techniques, and in adopting loss functions better suited to the class-imbalanced segmentation problem.

# References


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4. https://www.youtube.com/watch?v=jO6cqFAF_XA&list=LL&index=10&t=71s
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